Files for Laser Cutting and Engraving

The laser cutter works from vector graphics, for example Adobe Illustrator, CorelDRAW, AutoCAD, or Rhino.

There are also useful free programs, for example Inkscape and 123D.

If you draw in a 3D program, for example SolidWorks, be aware that you need a 2D drawing (top view), or a sketch from Fusion.

Check that there are no lines in the drawing that should not be used. In CAD programs, construction lines are often included.

Lines that are to be engraved or cut through may be placed on different layers, but this is not required. Check that there are not several identical lines lying on top of each other. Check that geometries are joined at the corner points if they are to be engraved.

Digital black-and-white images saved as JPG files can be engraved. However, it usually takes many tests before you get a good result. The result is best when there is high contrast in the image. A wire-haired pointer in a forest floor scene is impossible to make give a good result.

Make drawings at the size they are to be cut.

Illustrator

Draw at 1:1 scale - that is, at the size the finished item should have.

Line engraving does not use the stroke width specified in Illustrator.

For lines that are to be cut through, the specified stroke width is irrelevant.

Convert text to outlines. Do this by selecting the text box with the Selection Tool, right-clicking, and choosing **Create Outlines**.

File format: Illustrator (.ai) version CS5 or older, or PDF.

Rhino

Convert text to outlines.

Export Rhino files to Illustrator format (.ai).

Inkscape

Convert text to graphics (**Path menu - Object to Path**).

Save files as Inkscape .svg (standard format).

You should set the stroke width to **Hairline**.

SketchUp

Note that the free version does not export files at the correct scale.

Convert text to outlines.

Save files as .dxf.

When You Draw the File Yourself

Files for laser cutting can be drawn in Inkscape, Adobe Illustrator, CorelDRAW, or any CAD program, as long as the program draws in vectors. If you draw in a 3D program, remember that a 2D file is needed - top view.

Cutting Lines

For lines that are to be cut, choose the thinnest stroke width.

Lines that are not to be cut all the way through the material should be drawn in a different color.

Areas that are to be engraved should be drawn in a third color.

Be aware of whether any lines are lying on top of each other.

Engraving

Black-and-white images and fills will be engraved.

Lines that are to be engraved should be converted to objects.

Digital black-and-white images saved as JPG files can be engraved. The result is best when there is high contrast in the image. Make drawings at the size they are to be cut.

Check the File

All text for cutting or engraving must be vectorized (Create Outline, Create Contour, Object to Path, etc., depending on the program).

Make sure there are no lines in the drawing that should not be cut.

Check that there are not several identical lines lying on top of each other.

If the figure is to be cut or engraved, remember that the lines must be connected - also at the corners.

- Check that nothing is included that should not be cut or engraved.

Layers that should not be cut must be deleted - it is not enough to turn them off.

Do not use multiple pages in the document.

If a line needs to be cut around the object, there must be an actual line - it is not enough that the document is set up to the outer dimensions.